

Optimization

3.3.2 Newton's method

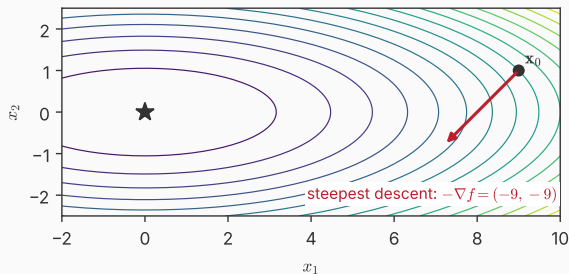
José Firmino Aguilar Madeira

Can we use the curvature?

The question left by 3.3.1. The same quadratic: $f(\mathbf{x}) = \frac{1}{2}(x_1^2 + 9x_2^2)$, $\mathbf{x}_0 = (9, 1)$,

$$\nabla f(\mathbf{x}) = \begin{pmatrix} x_1 \\ 9x_2 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & 9 \end{pmatrix}, \quad \nabla f_0 = (9, 9).$$

Steepest descent: $\mathbf{S} = -\nabla f_0 = (-9, -9)$
points at the “wall”: 74 iterations.



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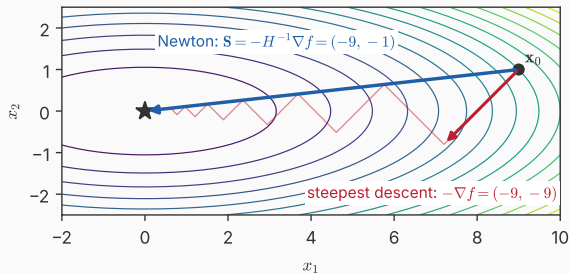
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Newton: solve $H\mathbf{S} = -\nabla f_0$:

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One iteration.



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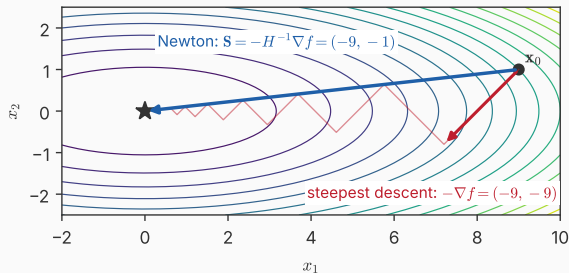
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Why does Newton “know” how to go straight to the minimum?

Where does Newton come from? The Taylor quadratic

At $\mathbf{x}_k$, approximate f by its Taylor quadratic (as in 2.5, now in $\mathbb{R}^n$):

$$f(\mathbf{x}_k + \mathbf{s}) \approx q(\mathbf{s}) = f(\mathbf{x}_k) + \nabla f_k^T \mathbf{s} + \frac{1}{2} \mathbf{s}^T H_k \mathbf{s}, \quad H_k = \nabla^2 f(\mathbf{x}_k).$$

We look for the minimum of the **model**, not directly of f : $\nabla q(\mathbf{s}) = \nabla f_k + H_k \mathbf{s} = \mathbf{0}$, hence

$$\boxed{H_k \mathbf{S}_k = -\nabla f_k} \quad (\text{formally } \mathbf{S}_k = -H_k^{-1} \nabla f_k), \quad \mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{S}_k.$$

- On the computer H_k^{-1} is **never computed**: the system $H_k \mathbf{S}_k = -\nabla f_k$ is solved (Cholesky, Gaussian elimination).
- If f is quadratic with $H \succ 0$, the model is *exact*: the minimum of the model is the minimum of f — **one iteration**, whatever $\mathbf{x}_0$. This is what we saw on the previous page.
- In general the model is only good near $\mathbf{x}_k$: hence the questions on the following pages.

Steepest descent vs. Newton: slope vs. curvature

	Steepest descent (3.3.1)	Newton (3.3.2)
uses	∇f	∇f and H
direction	$\mathbf{S} = -\nabla f$	$H\mathbf{S} = -\nabla f$
geometry	slope	slope + curvature
SPD quadratic	zigzag (linear)	1 iteration
cost per iteration	low	high (Hessian + system)

From 3.3 (“From 3.1. . .”): $\mathbf{S} = -M\nabla f$ with $M \succ 0$ is a descent direction; steepest descent uses $M = I$; when $H \succ 0$, Newton uses $M = H^{-1} \succ 0$.

H^{-1} rescales and rotates the gradient according to the curvature.

In the example: $H^{-1} = \text{diag}(1, \frac{1}{9})$ shrinks the x_2 component (the direction of strong curvature) nine times: $(-9, -9) \rightarrow (-9, -1)$.

When is the Newton direction a descent direction?

$$\nabla f_k^T \mathbf{S}_k = -\nabla f_k^T H_k^{-1} \nabla f_k.$$

- If $H_k \succ 0$ (then $H_k^{-1} \succ 0$): $\nabla f_k^T \mathbf{S}_k < 0$ for every $\nabla f_k \neq \mathbf{0}$ — descent guaranteed.
- If H_k is indefinite or negative definite: there is no guarantee of descent.
- Near a **nondegenerate** strict minimum ($H(\mathbf{x}^*) \succ 0$), by continuity $H_k \succ 0$: the guarantee is local.
(A strict minimum can have singular $H(\mathbf{x}^*)$: $f(x) = x^4$ at 0.)

What can actually happen when H_k is not positive definite? — shortly, on Himmelblau.

Newton's algorithm

Newton: $\mathbf{x}_0, \varepsilon$

- Step 1** Choose $\mathbf{x}_0$ and $\varepsilon > 0$; set $k = 0$.
- Step 2** Newton direction: set $\mathbf{S}_k = -H_k^{-1}\nabla f_k$,
with $H_k = \nabla^2 f(\mathbf{x}_k)$
(computed by solving $H_k \mathbf{S}_k = -\nabla f_k$).
- Step 3** Optimal step length: λ_k^* minimizes
 $f(\mathbf{x}_k + \lambda \mathbf{S}_k)$, $\lambda > 0$.
- Step 4** Set $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k^* \mathbf{S}_k$.
- Step 5** If $\|\nabla f_{k+1}\| \leq \varepsilon$, **stop**.
Otherwise, set $k = k + 1$ and go back to step 2.

- It is steepest descent with **another direction** in step 2.
- For an SPD quadratic, $\lambda_k^* = 1$ (next page): one iteration, from any $\mathbf{x}_0$.
- **Pure Newton**: in step 3, set $\lambda_k = 1$ with no line search — the model itself says how far to go. On quadratics it is the same; elsewhere it may increase f (Rosenbrock, next).
- **Local quadratic** convergence, under regularity and with $H(\mathbf{x}^*) \succ 0$.

Why is $\lambda_k^* = 1$ on a quadratic?

$f = \frac{1}{2}\mathbf{x}^T A \mathbf{x} + B\mathbf{x} + C$ with A SPD: $\nabla^2 f = A$ at every point and $\mathbf{S}_k = -A^{-1}\nabla f_k$. By the 2nd method for the optimal step (3.3.1):

$$\lambda_k^* = \frac{-\nabla f_k^T \mathbf{S}_k}{\mathbf{S}_k^T A \mathbf{S}_k} = \frac{\nabla f_k^T A^{-1} \nabla f_k}{\nabla f_k^T A^{-1} A A^{-1} \nabla f_k} = 1.$$

The intuition

Newton minimizes the 2nd-order Taylor model. On a quadratic the model **is** f itself: $\mathbf{x}_k + \mathbf{S}_k$ is already the minimum of f , hence also the minimum along the line — at $\lambda = 1$.

Class example:

$f = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$, $\mathbf{x}_0 = (0; 0)$.

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}, \quad \nabla f_0 = (1; -1),$$

$$\mathbf{S}_0 = -A^{-1}\nabla f_0 = (-1; \frac{3}{2}).$$

$$-\nabla f_0^T \mathbf{S}_0 = \frac{5}{2}, \quad \mathbf{S}_0^T A \mathbf{S}_0 = \frac{5}{2} \quad \Rightarrow \quad \lambda_0^* = 1.$$

$\mathbf{x}_1 = (-1; \frac{3}{2})$: the minimum, in one iteration.

And if f is not quadratic? Computing λ_k^*

$$f(\mathbf{x}) = x_1^2 + x_2^4, \mathbf{x}_0 = (1; 1), f(\mathbf{x}_0) = 2.$$

$$\nabla f = (2x_1; 4x_2^3), H = \text{diag}(2; 12x_2^2): \nabla f_0 = (2; 4), \\ H_0 = \text{diag}(2; 12).$$

Step 2. $\mathbf{S}_0 = -H_0^{-1}\nabla f_0 = (-1; -\frac{1}{3}).$

Step 3. $f(\mathbf{x}_0 + \lambda\mathbf{S}_0) = (1 - \lambda)^2 + (1 - \frac{\lambda}{3})^4$ is **not a parabola** in λ :

$$\frac{df}{d\lambda} = -2(1 - \lambda) - \frac{4}{3}\left(1 - \frac{\lambda}{3}\right)^3 = 0$$

is a cubic equation. Solve it with a method of Ch. 2, for example Newton–Raphson (2.5) applied to $df/d\lambda$, from

$$\lambda = 1, \text{ with } \frac{d^2f}{d\lambda^2} = 2 + \frac{4}{3}\left(1 - \frac{\lambda}{3}\right)^2:$$

$$\lambda = 1 - \frac{-32/81}{70/27} = 1 + \frac{16}{105} = 1.1524 \rightarrow 1.1551 = \lambda_0^*.$$

	$\mathbf{x}_1$	$f(\mathbf{x}_1)$
$\lambda = 1$ (pure)	(0; 0.667)	0.1975
$\lambda_0^* = 1.155$	(-0.155; 0.615)	0.1671

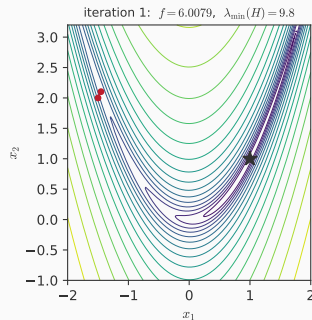
Caution

The 2nd-method formula with H_0 in place of A gives $\frac{-\nabla f_0^T \mathbf{S}_0}{\mathbf{S}_0^T H_0 \mathbf{S}_0} = \frac{10/3}{10/3} = 1$: it is the λ of the **model**, not that of f . Outside quadratics, λ_k^* is computed by the **1st method** ($df/d\lambda = 0$, or a line search from Ch. 2).

Rosenbrock with pure Newton: very fast, but not monotone

Pure Newton ($\lambda_k = 1$) from $(-1.5, 2)$.

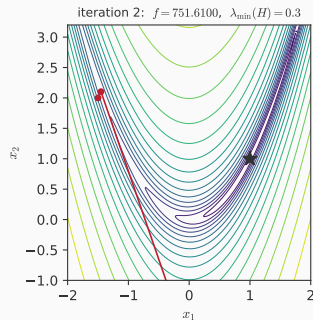
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0	$(-1.5000, 2.0000)$	12.500000	$1.63 \cdot 10^2$
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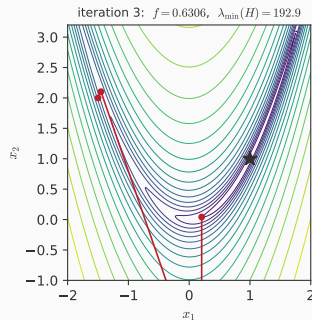
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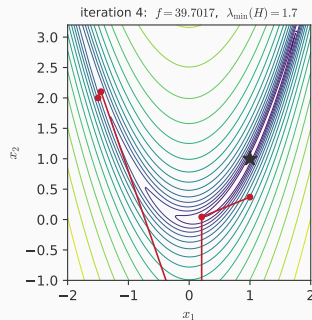
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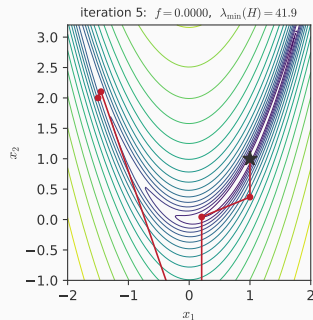
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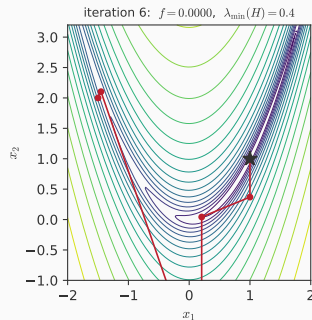
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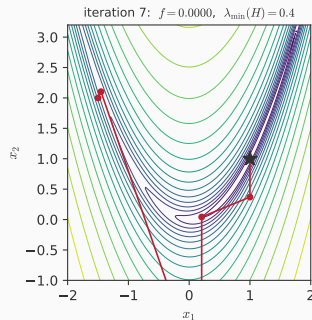


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7	$(1.0000, 1.0000)$	0	0

Along the way, f rises to 751 and to 40: the full step is **not monotone** — and even so it converges, with quadratic acceleration at the end. With λ_k^* (the class algorithm), at $k = 1$: $\lambda_1^* = 0.133$ and $f = 5.23$ instead of 751.6.

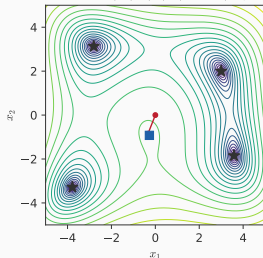


Steepest descent: 3000 it., still $f \approx 2.4 \cdot 10^{-4}$.
Pure Newton: **7 iterations**, $f < 10^{-16}$.

But comparing iterations is not enough: one Newton iteration is much more expensive (Hessian + system).

Newton seeks $\nabla f = 0$ — not necessarily minima

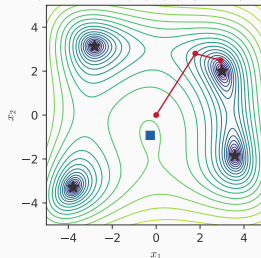
Pure Newton from $(0,0)$: $\lambda(H_0) = (-42, -26) \rightarrow f = 181.617$



Himmelblau from $(0,0)$: $\lambda(H_0) = (-42, -26)$, H_0 **negative definite**. The Newton direction points **uphill**; the pure method converges in 4 iterations — to the **local maximum** $(-0.27, -0.92)$, $f = 181.6$.

The damped method detects $\nabla f^T \mathbf{S} \geq 0$, uses $-\nabla f$ with line search and reaches the minimum $(3, 2)$ in 5 iterations.

Damped Newton from $(0,0)$: $\lambda(H_0) = (-42, -26) \rightarrow f = 0.000$



Newton seeks $\nabla f(\mathbf{x}) = 0$

Pure Newton solves the first-order conditions — and $\nabla f = 0$ includes **minima, maxima and saddles** (3.1: stationary $\neq$ minimum). Only if $H_k \succ 0$: *unique* model minimum and a descent direction.

Damped Newton: protect when far, let it run when near

1. compute $\nabla f_k, H_k$;
2. test $H_k \succ 0$ (Cholesky);
3. if it fails: modify, $H_k + \mu I$ with sufficiently large $\mu > 0$, or use $-\nabla f_k$;
4. solve $H_k \mathbf{S}_k = -\nabla f_k$;
5. ensure $\nabla f_k^T \mathbf{S}_k < 0$ (otherwise, $\mathbf{S}_k = -\nabla f_k$);
6. step length: exact λ_k^* (as in class) or only sufficient decrease (Armijo/Wolfe, next), starting at $\lambda = 1$; $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{S}_k$.

far: protect

near: $\lambda_k \rightarrow 1$, let Newton run

Property (globalized Newton)

With an Armijo/Wolfe-type line search starting at $\lambda = 1$, under suitable conditions, once the iterates are sufficiently close to the solution the full step $\lambda_k = 1$ is accepted — and quadratic convergence is recovered.

In the experiments of this course (safeguard of step 5 + very accurate one-dimensional minimization): Rosenbrock, pure 7 iterations (not monotone), damped 14 (monotone); the last line minimizers were 0.91, 0.98, 1.00 — a numerical illustration consistent with the property, not a proof. On Himmelblau, the damped method avoids the local maximum.

Which λ to accept? The Armijo condition

Minimizing $\varphi(\lambda)$ accurately is expensive: on Rosenbrock, the damped method of the previous slide used **about 360 evaluations of f** in 14 iterations. In practice a **sufficient decrease** is enough.

Armijo condition

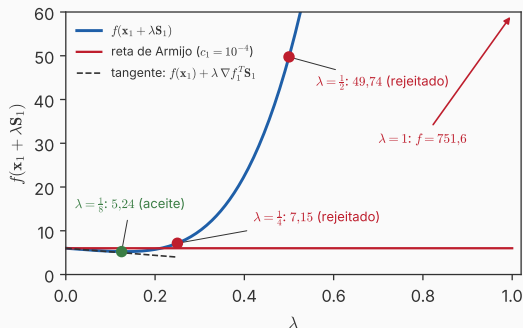
$$f(\mathbf{x}_k + \lambda \mathbf{S}_k) \leq f(\mathbf{x}_k) + c_1 \lambda \nabla f_k^T \mathbf{S}_k, \quad 0 < c_1 < 1$$

(typically $c_1 = 10^{-4}$)

Backtracking: $\lambda = 1, \frac{1}{2}, \frac{1}{4}, \dots$ until it holds.
Starting at $\lambda = 1$ lets Newton run.

At $k = 1$: $f(\mathbf{x}_1) = 6.008$, $\nabla f_1^T \mathbf{S}_1 = -8.12$; $\lambda = 1, \frac{1}{2}, \frac{1}{4}$ rejected; $\lambda = \frac{1}{8}$ **accepted** (4 evaluations).

Rosenbrock with Armijo: **22 iterations**, $n_f = 29$, f always decreasing; $\lambda = 1$ in the last 7. More iterations, far fewer evaluations.



Rosenbrock, $\mathbf{x}_1 = (-1.451, 2.103)$ (pure Newton), Newton direction $\mathbf{S}_1$.
The Armijo line is almost horizontal: a small decrease is enough.

Wolfe adds a condition on $\varphi'(\lambda)$ that rules out steps that are too short (essential in BFGS).

What if H_k is not positive definite?

1. Change the direction

$\mathbf{S}_k = -\nabla f_k$ when $\nabla f_k^T \mathbf{S}_k \geq 0$.

Simple; the curvature is lost in that iteration. This is what was done on Himmelblau.

2. Modify the Hessian

$(H_k + \mu I)\mathbf{S}_k = -\nabla f_k$, with $\mu > 0$ until Cholesky succeeds.

Small μ : almost Newton; large μ : $\mathbf{S}_k \approx -\nabla f_k / \mu$, almost steepest descent. (The same idea as Levenberg–Marquardt, in least squares.)

3. Trust regions

$\min_{\mathbf{s}} q(\mathbf{s})$ s.t. $\|\mathbf{s}\| \leq \Delta_k$.

In the ball the problem has a solution even with H_k indefinite.

$\rho_k = \frac{f(\mathbf{x}_k) - f(\mathbf{x}_k + \mathbf{s})}{q(\mathbf{0}) - q(\mathbf{s})}$: large $\Rightarrow$ accept and enlarge Δ ; small $\Rightarrow$ reject and shrink Δ .

Himmelblau from $(0, 0)$, $\lambda(H_0) = (-42, -26)$, with Armijo:

	iterations	n_f
1. $-\nabla f$ at the 1st iteration	6	14
2. $H + \mu I$	8	10

Both reach the minimum $(3, 2)$.

Line search: first the **direction**, then the length.

Trust region: first the **size** Δ_k , then the direction.

Trust regions: outside the syllabus (Nocedal–Wright, Ch. 4).

The price of Newton and what grows from it: quasi-Newton

Price per iteration: $n_g = 1$, $n_H = 1$ ($2n^2 + 1$ evaluations of f if numerical); dense system $O(n^3)$; memory $O(n^2)$. With millions of variables, classical Newton is not the natural choice.

Idea: approximate the curvature with the gradients we already have. With $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \nabla f_{k+1} - \nabla f_k$, require $B_{k+1} \mathbf{s}_k = \mathbf{y}_k$ (secant equation; in 1D, the secant line).

- **BFGS:** a simple update of B_k (or of its inverse) that keeps it positive definite if $\mathbf{y}_k^T \mathbf{s}_k > 0$ (Wolfe guarantees it); no second derivatives; **superlinear** convergence.
- **L-BFGS:** keeps only the last m pairs $(\mathbf{s}, \mathbf{y})$, $m \approx 5\text{--}20$: memory $O(mn)$.

method	memory	cost/it.*	local conv.
steepest descent (3.3.1)	$O(n)$	$O(n)$	linear
conj. gradients (3.3.3)	$O(n)$	$O(n)$	linear**
L-BFGS	$O(mn)$	$O(mn)$	linear
BFGS	$O(n^2)$	$O(n^2)$	superlinear
Newton (3.3.2)	$O(n^2)$	$O(n^3)$	quadratic

* besides the evaluations of f , ∇f (and H). ** n iterations on SPD quadratics (3.3.3).

In software: MATLAB `fminunc` ('quasi-newton', BFGS, by default; 'trust-region', with the gradient supplied); SciPy `minimize` ('BFGS', 'L-BFGS-B', 'trust-exact', ...).

Quasi-Newton and trust regions: **outside the syllabus** — this is where the subject continues.

- Solve $HS = -\nabla f$ with $H \setminus g$, never with $\text{inv}(H)$.
- Test $H \succ 0$ with Cholesky: $[\sim, p] = \text{chol}(H)$, $p = 0$ if $H \succ 0$.
- The line search of the companion code: line minimizer (Brent, by default) or Armijo backtracking (`opts.ls = 'armijo'`); $H + \mu I$ with `opts.modify = true`.

Newton: **exact** curvature, expensive

BFGS: **approximate** curvature

The core, in MATLAB:

```
g = grad(x);  H = hess(x);
S = -(H \ g);      % H*S = -g
if g'*S >= 0
    S = -g;        % safeguard
end
lam = ...;        % line search
x = x + lam*S;
```

The full function (NewtonND, pure/damped, with `info.nfev`, `ngev`, `nhev`) is in the course companion code.

KEY TAKEAWAYS — 3.3.2

1. Newton locally minimizes the Taylor quadratic: $H_k \mathbf{S}_k = -\nabla f_k$ (the system is solved, H_k is not inverted).
2. If f is an SPD quadratic, it reaches the minimum in **one iteration** ($\lambda_0^* = 1$).
3. Pure Newton seeks $\nabla f = \mathbf{0}$, **not minima**: if $H_k \not\succ 0$, the direction may not be a descent direction and the method may converge to maxima or saddles.
4. Optimal λ_k^* : 1 on an SPD quadratic; elsewhere, $df/d\lambda = 0$ or a line search. Pure Newton ($\lambda_k = 1$) is local; in practice: $H + \mu I$ and **sufficient decrease** (Armijo, starting at $\lambda = 1$). Near a minimum with $H(\mathbf{x}^*) \succ 0$, $\lambda_k \rightarrow 1$ and **quadratic** convergence is recovered (under regularity).
5. The price: Hessian ($O(n^2)$ memory) + system ($O(n^3)$). This is where **quasi-Newton** methods (BFGS: curvature from gradients) and **trust regions** come from.

Next (3.3.3): **can we avoid the zigzag of steepest descent without computing or storing the Hessian?**
— conjugate gradients.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Chs. 3 (line search, Armijo/Wolfe), 4 (trust regions), 6 (quasi-Newton) and 7 (L-BFGS).
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996 — Chs. 6 and 9 (secant).
- Conn, A. R., Gould, N. I. M., Toint, Ph. L., *Trust-Region Methods*, SIAM (MPS-SIAM Series on Optimization 1), 2000.
- Armijo, L., *Minimization of functions having Lipschitz continuous first partial derivatives*, Pacific Journal of Mathematics 16(1):1–3, 1966.

Resources

- Companion code: NewtonND (pure/damped; Brent or Armijo; $H + \mu I$; `info.nfev`, `ngev`, `nhev`).
- MATLAB: `chol`; `fminunc` (BFGS by default; `'trust-region'`). SciPy: `minimize`.